Responsible AI Framework: Project Design Considerations

NovaCare Health Systems

Arshi Khan

Assignment: 5.4 RAI Framework Project: Lifecycle and Sustainment

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NovaCare Health Systems

Fictional healthcare enterprise (consistent with Assignments 1.3 and 1.4)

- **Scale** — 30 hospitals, 150+ outpatient facilities, approximately 45,000 employees
- **Reach** — Serves about 6 million patients annually through hospitals, clinics, diagnostic centers, and telehealth
- **Technology footprint** — Enterprise EHR, cloud analytics, imaging, patient portals, and digital care platforms
- **RAI foundation** — Seven ethical principles established in 1.3, operationalized through the NIST AI RMF (Govern, Map, Measure, Manage) in 1.4

AI Use Cases Under Consideration

Same four use cases established in Assignment 1.3

1

AI-Assisted Medical Imaging

Detects abnormalities in X-rays, CT scans, and MRIs

2

Patient Deterioration Prediction

Predicts patients at risk of rapid clinical decline

3

Generative AI Clinical Assistant

Summarizes records and drafts clinical documentation

4

AI Scheduling & Resource Optimization

Forecasts demand and optimizes staffing and appointments

Chart: Full AI Development Lifecycle and RAI Actions

Extends the 7-step vetting process from Assignment 1.4 with sustainment and disposition stages

Stage	RAI Action	Threshold / Trigger
1. Design & Scoping	Use-case charter completed and risk-tiered before any data work begins	*No data collection starts without a signed-off charter*
2. Data Collection & Vetting	Subgroup representation audit; documented data lineage	*No dataset advances without documented lineage*
3. Development & Training	Model card drafted in parallel with training, not after	*Model card completion is a training-sprint exit criterion*
4. Testing & Validation	Independent validation against fairness, safety, and explainability gates	*Cannot proceed to deployment without passing all gates*
5. Deployment	Controlled, phased rollout with human oversight active from day one	*Override/complaint channel live before first patient-facing use*
6. Continuous Monitoring	Track drift, subgroup performance, and incidents monthly	*Automatic flag if accuracy drops more than 5 percentage points from baseline*
7. Sustainment (Retrain/Tune)	Scheduled retraining as data or patient population shifts	*Mandatory retrain if drift exceeds threshold for 2 consecutive months, or annually regardless*
8. Disposition/Replacement	Formal retirement when a system no longer meets its safety/fairness bar	*Retirement criteria set at approval time; replacement must clear the same validation gates first*

Chart: Commercially Available RAI Validation/Monitoring Tools

Use Case	Tool	Why It Fits
AI-Assisted Medical Imaging	Fiddler AI	*Enterprise-grade explainability and compliance-focused monitoring, suited to regulated diagnostic use*
Patient Deterioration Prediction	IBM Watson OpenScale	*Built for regulated industries; bias, fairness, and drift monitoring with healthcare-relevant governance*
Generative AI Clinical Assistant	Arize AI	*LLM tracing and evaluation; detects hallucination-prone outputs and embedding drift in generated text*
AI Scheduling & Resource Optimization	WhyLabs	*Lightweight data drift and model health monitoring; SOC 2 Type 2 compliant, cost-effective for a lower-risk tool*

Chart: Building RAI Expertise Within NovaCare

Strategy	Action	Measurable Target
Formal Training	Mandatory RAI onboarding module for all AI-adjacent staff	*100% completion within 30 days of hire; refresher every 12 months*
Role-Specific Certification	RAI role-holders (per 1.4's 7 roles) pursue relevant certifications (e.g., NIST AI RMF training)	*At least 1 certified staff member per RAI role within year 1*
Cross-Functional Rotation	Clinical staff periodically shadow the RAI Governance & Ethics Lead	*2 clinical staff per year participate*
Internal Community of Practice	Monthly RAI case-review forum open to all staff	*Minimum 20 attendees per month target*

Chart: Educating Customers on Appropriate AI Use

Audience	Method	Measurable Goal
Patients	Plain-language disclosure at point of care when AI contributes to a diagnosis or scheduling decision	*100% of AI-influenced encounters include a disclosure notice*
Patients	Patient portal FAQ and short explainer video on how AI is used and patient rights	*Available in top 3 languages spoken by the patient population*
Referring Clinicians (external)	Onboarding guide for partner facilities using NovaCare's AI-assisted imaging referral network	*Guide acknowledgment required before system access is granted*
General Public	Annual transparency report summarizing AI use and outcomes, publicly posted	*Published annually, no exceptions*

Conclusion

Difficulties and open questions from this workshop

- The lifecycle chart makes retraining and disposition look like clean, scheduled steps, but figuring out how much drift actually counts as "too much" felt more like a judgment call than something you could measure precisely.
- The emotional support robot getting bricked stuck with me. It wasn't a technical failure, it was a business decision with no plan for the people who trusted the product when the company just walked away.
- I'm not sure NovaCare could realistically hit "retrain within 2 consecutive months of drift" in practice. It looks clean on a slide, but real retraining takes engineering time and budget that might not move that fast.

Open Question: who actually decides when a model has failed badly enough to be retired instead of just retrained? That felt like the least defined step in the whole lifecycle.

References

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